Joint Structure & Function.

Definition of joint:- A joint is a place where two or more bones come together to allow movement, provide support and maintain structural connection in the body.

Imp characteristics of Joints.

- Connection Point:- Joints connect ~~to~~ 2 or more bones to form the skeleton.
- Allow movement:- Many joints Allow free or limited movement depending on their type.
- Support & Stability:- Joints help maintain posture and body stability during standing and walking.
- Contain Soft structures:- Some joints contain cartilage, ligaments, tendons and fluid to reduce friction and absorb shock.
- Development & Growth:- In growing children, joints like epiphyseal plates help in bone lengthening.
- Protection of Organs:- Some joints like those in the skull and ribcage protect vital organs like the brain and heart.

Classification of Joints.

Joints are classified in two main ways:

1. Structural Classification (It is based on type of tissue & joint cavity)

Includes:
- Fibrous joints (Synarthroses)
- Cartilaginous joints (Amphiarthroses)
- Synovial joints (Diarthroses)

I Fibrous joints (Synarthroses / Articulationes Fibrosae).
- Bones are connected by dense fibrous ct.
- No joint cavity is present
- These joints allow little or no movement.

Types.
- Sutures.
  - Found in skull
  - Bones edges interlock
  - Immovable
  - eg:- Coronal suture (frontal and parietal bone)
- Syndesmoses
  - Connected by ligaments / Interosseous membrane
  - Slight movement possible
  - eg:- distal tibiofibular joint.
- Gomphoses.
  - Peg-in-socket joint
  - eg:- Tooth in socket (maxilla/mandible)

II Cartilaginous Joints (Amphiarthroses / Articulationes Cartilaginae)

- Bones are united by cartilage (either hyaline or fibrocar)
- No joint cavity is present.
- Allows limited movement.
- ~~eg - Pubic symphysis, intervertebral discs.~~

Types:

- Synchondroses (Primary Cartilaginous joints)
  - Bones joined by hyaline cartilage.
  - Usually temporary, ossify later.
  - Movement is absent.
  - eg:- Epiphyseal plate, Ist sternocostal joint.

- Symphyses (Secondary Cartilaginous joints)
  - Bones joined by fibrocartilage
  - Permanent joints
  - Allow limited movement.
  - eg:- Pubic symphysis, intervertebral discs.

III Synovial Joints (Diarthroses / Articulationes Synoviales)
- Most common and movable joints in the body.
- Bones are separated by a joint cavity filled with synovial fluid
- Allow free movement.

Structural Components:

- Joint Capsules (Capsula articularis)
  - Outer fibrous layer + inner synovial membrane.

- Synovial membrane
  - Produces synovial fluid.

- Synovial fluid
  - Lubricates and nourishes the joint.

- Articular cartilage
- Covers bone ends, made of hyaline cartilage.

- Accessory structures:
- Ligaments, bursae, tendons, menisci, etc.

Examples:
- Shoulder joint (Glenohumeral joint)
- Knee joint (Tibiofemoral joint)
- Hip joint (Acetabulofemoral joint)

Main Structural Features of Synovial Joints:
- Joint Capsule (Capsula Articularis):
- A tough, fibrous outer covering that surrounds the entire joint.
- It provides protection and stability to the joint.

- Synovial membrane:
- The inner layer of the capsule.
- It secretes synovial fluid.

- Synovial Fluid:
- A thick, slippery fluid that lubricates the joint, reduces friction and nourishes the cartilage.

- Articular Cartilage
- A smooth layer of hyaline cartilage that covers the ends of bones.
- It prevents bones from rubbing against each other

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- Pronation - Supination.
- Pronation: Turning the palm downward.
- Supination: Turning the palm upward.

Types of Synovial Joints & their Working
- Synovial joints are classified based on the shape of the articulating surfaces and the type of movement they allow.

1. Ball and Socket Joint.
Structure: One bone has a ball-shaped head that fits into a cup-like socket of another bone.

Movements:
- Flexion & Extension
- Abduction & Adduction.
- Rotation
- Circumduction

Example:
- Shoulder Joint - Humerus + Scapula.
- Hip Joint - Femur + Pelvis

2. Hinge Joint.
structure: Convex Surface of One bone fits into concave surface of another.

Movements:
- Flexion & Extension only.

Example:
- Elbow joint - Humerus + Ulna.

- Knee Joint - Femur + Tibia
- Interphalangeal Joints - Finger joints

3. Pivot Joint
Structure: A ring-like structure rotates around a peg or rounded bone.

Movements.
- Rotation only.

Example:
- Atlantoaxial joint - Between atlas (C1) and axis (C2) vertebrae (used when you turn your head left/right)

- Radioulnar joint - Allows forearm rotation.

4. Condyloid Joint (Ellipsoidal Joint)
Structure: Oval-shaped bone fits into elliptical cavity of another.

Movements:
- Flexion & Extension
- Abduction & Adduction
- No rotation.

Example:
- Wrist Joint - Radius + Carpal bones
- Metacarpophalangeal joints - knuckle joints.

5. Saddle Joint.

Structure: Each bone has both concave and convex surfaces, fitting like a rider on a saddle.

Movements:
- Flexion & Extension
- Abduction & Adduction
- Limited Circulation.

Example:
- Thumb joint - Allows opposition.

6. Plane (Gliding) Joint.
Structure: Flat or slightly curved surfaces slide past each other.

Movements:
- Gliding or Sliding movements only.

Example:
- Intercarpal joints - Between wrist bones
- Intertarsal joints - Between ankle bones
- Acromioclavicular joint - Shoulder region.

1. Joint Motion:
In simple term, it is the movement that occurs at the place where two or more bones connect, allowing for the body to bend, twist, and move in different directions.

- Joint motion means movement between two bones.

- Range of Motion (ROM) is the full movement possible at a joint.
- ROM depends on joint shape, capsule, ligaments and muscles.
- Eg: Elbow bends and straightens easily.

- Too much ROM is called hypermobility.
- Very little ROM is called hypomobility.
- Movements happen in planes like sagittal, frontal & transverse.
- Joints can be uniaxial, biaxial or triaxial (eg:- elbow, wrist, shoulder respectively).
- ROM is measured using goniometer (special device for measuring joint angles).
- Regular stretching exercises help to improve ROM.

2. Arthrokinematics (Joint Surface movements)
- Arthrokinematics means small movements at the joint surface.
- It includes three types: roll, slide and spin.
- eg: i) Roll: like a ball rolling on the floor.
ii) Slide: like a car tyre slipping on wet road.
iii) Spin: like a top spinning on one point.
- These happen naturally during normal joint movement.
- Roll and glide must happen together for smooth joint function.
- Poor arthrokinematics can cause joint pain and stiffness.
- Therapists use specific techniques to restore proper roll, slide and spin after injuries.

- It is safest for healing, after injury or during light exercises.
- Physiotherapists uses loose-packed position to apply gentle mobilizations.
- Less joint congruency.
- Loosed-packed is the best position for resting painful joints.

1. Degree of Freedom (DOF)

Definition:

- Degree of freedom refers to the number of independent movements a joint can perform.
- Every joint movement happens in a specific plane and around a specific axis.

Explanation:

- More degree of freedom = More mobility.
- less degree of freedom = More stability.
- The structure of the joint decides how many DOF it will have.

Types of Degree of Freedom:

A) 1 Degree of freedom (1 DOF).

- Joint moves in one plane only.
- Plane: Sagittal plane
- Axis: Frontal axis
- Joint type: Hinge joint.
- eg: i) Elbow joint
  ii) knee joint
- Daily life examples:- Eating food, lifting b
- Special point :- Very stable but limited motion.

in space without any restriction.
- Movements occur at a single joint without affecting adjacent joints.
- There is no weight bearing on the limb during movement.
- Exercises focus on isolating individual muscles for strengthening.
- OKC movements are generally non-functional.
- Resistance is applied directly to the moving part.
- eg: of OKC include leg extension, seated bicep curl and kicking a ball.

Closed kinetic chain (CKC)
- In CKC, the distal segment is fixed and cannot move freely.
- Movements occur at multiple joints simultaneously
- CKC movements are always weight-bearing.
- Exercises improve joint stability, balance and co-ordination.
- CKC activities are generally functional movements.
- Resistance is distributed across multiple joints.
- Examples of CKC include squats, push-up, and standing up from a chair.

Feature	Osteokinematics	Arthrokinematics
Meaning.	Movement of bones around a joint axis	Movement between joint surfaces.
Type of movement	Large, visible movements.	Small, hidden joint movements.
Movements involved	Flexion, Extension, Abduction Rotation	Roll, Glide (slide), spin.
Observation	Can be seen directly (eg. lifting arm).	Cannot be seen directly.
Measurement tool	Goniometer	Special tests/manual feeling.
Planes and axes	Moves in planes (sagittal, frontal, transverse)	Occurs inside joint (no fixed plane).
Clinical Importance.	Range of Motion (ROM) assessment	Joint stability and integrity maintenance

(B) 2 Degrees of Freedom (2 DOF)
- Joint moves in two planes.
- Planes :- i) Sagittal plane
  ii) Frontal plane.
- Axis : Frontal and Sagittal axis
- Joint type: Condyloid joint.
- Eg : i) wrist joint.
  ii) Ankle joint.
- Daily life eg :- Writing, turning door knob.
- Special point: Balanced mobility and stability.

(C) 3 Degrees of Freedom (3 DOF)
- Joint moves in 3 planes.
- Planes :- i) Sagittal
  ii) Frontal
  iii) Transverse
- Axis : Frontal, Sagittal, Vertical axis.
- Joint type: Ball & socket Joint.
- eg :- i) shoulder joint.
  ii) Hip joint
- Daily life eg :- Throwing ball, swimming.
- Special point: Highly mobile but less stable.

2. Kinetic Chains.
- Kinetic chain refers to how joints and muscles work together during movement.
- It is of two types : Open kinetic chain (OKC) and Closed kinetic chain (CKC).

Open kinetic chain (OKC)
- . In OKC, the distal segment moves freely

- Poor stability leads to loose joints and frequent ~~joints~~ injuries.
- Stability exercises like resistance band work can strengthen joints.

5. Close-Packed Position.
- In close-packed position, joint surfaces are tightly fit and capsule - ligaments are very tight.
- eg: i) knee fully extended is close-packed
  ii) shoulder fully extended rotated is close-packed.
- This position is very stable but injury risk is higher if force is applied.
- Used during diagnosis to test ligament and capsule strength.
- Maximum joint congruency occurs in close-packed position.
- Weight-bearing joints like hip show close-packed position during standing.
- Avoid overloading joints in close-packed position to prevent injury.

6. Loose-Packed Position.
- In loose-packed position joint surfaces are slightly apart and capsule - ligaments are relaxed.
- Joint space is maximum here, reducing internal pressure.
- eg: i) knee half bent is loose-packed
  ii) shoulder at mid-range is loose-packed.

3. Concave-Convex Rule.
- when a Concave surface moves, roll and glide are in the same direction.
- when a convex surface moves, roll and glide are in opposite directions.
- eg : i) Shoulder joint : Ball (humerus head) is convex and moves opposite to arm motion.
ii) Knee joint : Tibia (concave) moves in the same direction as the foot.
- This rule is important for joint mobilization during physiotherapy.
- Helps to apply correct force during treatment.
- Wrong glide direction can increase pain and cause damage.
- Learning this rule helps physiotherapists give better joint exercises.

4. Joint Stability.
- Stability means how well a joint can hold together under force.
- It depends on joint shape, ligaments, capsule and muscle control.
- Static stabilizers : Ligaments, capsule, labrum (cartilage pad).
- Dynamic stabilizers : Muscles that move and control joints.
- Eg : shoulder joint is stabilized by rotator cuff muscles.
- Good stability prevents dislocation and injury.
- Joint Compression naturally increases stability.

- Joint Cavity:
- A space between the bones that contains synovial fluid, allowing free movement.

- Accessory Structures:
- Ligaments: Connect bone to bone and provide stability
- Tendons: Connect muscles to bones
- Bursae: Small fluid-filled sacs that reduce pressure & friction
- Menisci: Crescent-shaped cartilage for shock absorption (eg: in the knee).

Types of Movements in Synovial Joints:
- Flexion - Extension
- Flexion: Decreasing the angle between two bones.
- Extension: Increasing the angle.

- Abduction - Adduction
- Abduction: Moving a limb away from the body
- Adduction: Moving it towards the body.

- Rotation
- Bone moves around its axis (eg:- turning the head side to side).

- Circumduction
- Circular movement (eg: making a circle with the arm).

- Gliding
- One bone slides over another (eg: small bones in the wrist).